Chenxi Liu

Assistant Professor, University of Victoria

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Research Interests

I am an Assistant Professor of Computer Science at the University of Victoria, where I lead the Creative eXperience Lab (CXL). My research aims to better understand and assist visual creation, currently focused on image generation, 2D visual representations such as vector graphics, and interactive tools for creative workflows.

Research & Professional Experience

Jul 2026 – Present	**Creative eXperience Lab (CXL), University of Victoria** Assistant Professor, Department of Computer Science.
Jul 2023 – Apr 2026	**DGP Lab, University of Toronto** Postdoctoral Fellow. Supervised by Prof. Alec Jacobson.
Mar 2023 – Jun 2023	**Adobe Research** Research Intern. Supervised by Deepali Aneja & Prof. Alec Jacobson.
May 2020 – Nov 2020	**Adobe Research** Research Intern. Supervised by Aaron Hertzmann.
Mar 2015 – Jul 2016	**Disney Research Pittsburgh** Research Associate. Supervised by Prof. James McCann & Prof. Jessica Hodgins.
Jul 2012 – Sep 2012	**Microsoft Search Technology Center Asia** Intern, Software Development Engineer in Test.

Education

Sep 2016 – Jun 2023	**University of British Columbia** Ph.D. in Computer Graphics. Advised by Prof. Alla Sheffer.
Aug 2013 – Dec 2014	**Carnegie Mellon University** M.S. in Computer Science.
Sep 2009 – Jun 2013	**Beihang University** B.Eng. in Computer Science and Technology. Ranking: 2/188.

Publications

Chenxi Liu, Selena Ling, and Alec Jacobson (July 2026). “GimmBO: Interactive Generative Image Model Merging via Bayesian Optimization”. In: *ACM Trans. Graph.* 45.4. Best Paper Award, SIGGRAPH North America 2026. ISSN: 0730-0301

Haiyang Xu, Ronghuan Wu, Li-Yi Wei, Nanxuan Zhao, **Chenxi Liu**, Cuong Nguyen, Zhuowen Tu, and Zhaowen Wang (2026). “SemLayer: Semantic-aware Generative Segmentation and Layer Construction for Abstract Icons”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 42082–42092

Yuta Noma, Zhecheng Wang, **Chenxi Liu**, Karan Singh, and Alec Jacobson (2026). “Mesh Processing Non-Meshes via Neural Displacement Fields”. In: *Computer Graphics Forum*

Chenxi Liu and Mikhail Bessmeltsev (2025). “State-of-the-art Report in Sketch Processing”. In: *Computer Graphics Forum*. Eurographics’25 STAR. Wiley Online Library

Chenxi Liu, Siqi Wang, Matthew Fisher, Deepali Aneja, and Alec Jacobson (2025). “2D Neural Fields with Learned Discontinuities”. In: *Computer Graphics Forum*. Eurographics’25. Wiley Online Library

Chenxi Liu, Towaki Takikawa, and Alec Jacobson (2024). *A LoRA is Worth a Thousand Pictures*. arXiv: 2412.12048 [cs.CV]

Siqi Wang, **Chenxi Liu**, Daniele Panozzo, Denis Zorin, and Alec Jacobson (2023). "Bézier Spline Simplification Using Locally Integrated Error Metrics". In: *SIGGRAPH Asia 2023 Conference Papers*. ACM

Chenxi Liu (2023). "Processing Freehand Vector Sketches". PhD thesis. University of British Columbia

Chenxi Liu, Toshiki Aoki, Mikhail Bessmeltsev, and Alla Sheffer (July 2023). "StripMaker: Perception-Driven Learned Vector Sketch Consolidation". In: *ACM Trans. Graph.* 42.4. SIGGRAPH'23

Chenxi Liu, Pierre Bénard, Aaron Hertzmann, and Shayan Hoshyari (Feb. 2023). "ConTesse: Accurate Occluding Contours for Subdivision Surfaces". In: *ACM Trans. Graph.* 42.1. SIGGRAPH'23

Jerry Yin, **Chenxi Liu**, Rebecca Lin, Nicholas Vining, Helge Rhodin, and Alla Sheffer (2022). "Detecting Viewer-Perceived Intended Vector Sketch Connectivity". In: *ACM Trans. Graph.* 41.4. SIGGRAPH'22

Dave Pagurek van Mossel, **Chenxi Liu**, Nicholas Vining, Mikhail Bessmeltsev, and Alla Sheffer (2021). "StrokeStrip: Joint Parameterization and Fitting of Stroke Clusters". In: *ACM Trans. Graph.* 40.4. SIGGRAPH'21

Yulia Gryaditskaya, Felix Hähnlein, **Chenxi Liu**, Alla Sheffer, and Adrien Bousseau (2020). "Lifting Freehand Concept Sketches into 3D". in: *ACM Trans. Graph.* 39.6. SIGGRAPH Asia'20, pp. 1–16

Chenxi Liu, Enrique Rosales, and Alla Sheffer (2018). "StrokeAggregator: Consolidating Raw Sketches into Artist-Intended Curve Drawings". In: *ACM Trans. Graph.* 37.4. SIGGRAPH'18

Chenxi Liu, Jessica Hodgins, and James McCann (July 2017). "Whole-cloth Quilting Patterns from Photographs". In: *Proceedings of the Symposium on Non-Photorealistic Animation and Rendering*. NPAR'17

Lea Albaugh, April Grow, **Chenxi Liu**, James McCann, Gillian Smith, and Jennifer Mankoff (2016). "Threadsteading: Playful Interaction for Textile Fabrication Devices". In: *Proceedings of the 2016 CHI Conference Extended Abstracts*. CHI'16. ACM

Honors

Jul 2026	**Best Paper Award: GimmBO** SIGGRAPH North America 2026.
Apr 2024	**Eurographics 2024 PhD Thesis Award, Honorable Mention** Eurographics and the Computer Graphics Forum Journal.
Feb 2024	**Faculty of Arts & Science Postdoctoral Fellowship** Faculty of Arts & Science, University of Toronto.
May 2022	**WiGRAPH Rising Stars** The ACM Community Group for Women in Computer Graphics Research.
Oct 2016	**Technology Award Winner: Threadsteading** IndieCade'16.
Nov 2011	**National Scholarship (Top 1% in Academic Performance)** Ministry of Education of the People's Republic of China.
Dec 2010 – 2012	**The First Prize Scholarship of Academic Performance** Beihang University.

Talks & Exhibition

October 2025 — **HiGen, ICCV 2025 Workshop**
Poster: Interactive T2I LoRA Merging with Human-in-the-Loop Bayesian Optimization.

October 2025 — **Concordia University, Université de Montréal, McGill University**
Invited Talk: From Representation to Workflow: Computational Tools for Human Visual Creativity.

August 2025 — **University of Victoria, Simon Fraser University**
Invited Talk: From Sketch to Style: Artistic Analysis and Creation via Computation.

May 2025 — **University College London, University of Surrey**
Invited Talk: From Sketch to Style: Analyzing Artworks Through Computation.

May 2025 — **Eurographics**
Conference Presenter: 2D Neural Fields with Learned Discontinuities.
State-of-the-art Report in Sketch Processing.

Apr 2025 — **Great Lakes Graphics Workshop@University of Chicago**
Invited Talk: 2D Neural Fields with Learned Discontinuities.

Jun 2024 — **ACM / Eurographics Symposium on Geometry Processing (SGP) Graduate School**
Course: Fundamentals and Applications of Sketch Processing.

Aug 2023 — **ACM SIGGRAPH**
Conference Presenter: ConTesse: Accurate Occluding Contours for Subdivision Surfaces.
StripMaker: Perception-driven Learned Vector Sketch Consolidation.

Fall 2022 — **University of Toronto, Université de Montréal, University of Surrey**
Invited Talk: Cleaning Up Vector Sketches Made by Humans and Computers.

Aug 2022 — **ACM SIGGRAPH**
Conference Presenter: Detecting Viewer-Perceived Intended Vector Sketch Connectivity.

Aug 2018 — **ACM SIGGRAPH**
Conference Presenter: Strokeaggregator: Consolidating raw sketches into artist-intended curve drawings.

Jul 2017 — **Non-Photorealistic Animation and Rendering (NPAR)**
Conference Presenter: Whole-cloth quilting patterns from photographs.

Mar 2016 — **Alt.Ctrl.GDC, Game Developers Conference**
Exhibitor: Threadsteading.
(Game also attended CHI Interactivity, 2016. IndieCade, 2016)

Teaching Experience

Fall, 2023 — **CSC317@UofT: Computer Graphics**
Guest Lecturer and Co-Developer of LEAF+ Project (Understanding the Limits of AI-Based Image Generators with DALL·E 2 and Midjourney). Project Lead: Prof. Alec Jacobson.

Winter 1, 2022 — **CPSC424@UBC: Geometric Modeling**
Graduate Teaching Assistant, Instructor: Prof. Alla Sheffer.

Winter, 2019 — **Instructional Skills Workshops for Grad Students@UBC**

Winter 2, 2018 — **CPSC436D@UBC: Video Game Programming**
Graduate Teaching Assistant, Instructor: Prof. Alla Sheffer.

Winter 2, 2016 — **CPSC418@UBC: Parallel Computation**
Graduate Teaching Assistant, Instructor: Prof. Mark Greenstreet.

Mentoring

2023 – 2024 | Sepehr Ghasemipour
University of Toronto.

2023 – 2024 | Silvia Lopez
University of Toronto.

2022 – 2023 | Toshiki Aoki
University of Tokyo.

Academic & Departmental Service

2026 | **Eurographics**
Diversity Panelist.

2025 | **SIGGRAPH Asia**
Technical Papers Committee.

2025 | **SIGGRAPH**
Birds of a Feather program co-organizer.
Countering Racial Bias in Computer Graphics Research,
Sustainable Research in Computer Graphics.

2025 | **Eurographics**
Diversity Panelist.

2025 | **Symposium on Geometry Processing**
International Program Committee.

2024 | **DGP Academy**
Project Lead: Low-Poly Fabrication. Media.

2019 – Present | **Conference and Journal Paper Review**
SIGGRAPH'22,23,24,25,26. SIGGRAPH Asia'22,24,25,26.
NeurIPS'26. BMVC'26. ICCV'25. WACV'25.
Eurographics'20,22,23,24,25. SGP'25. Pacific Graphics'20,24,25.
TVCG'23,24,25,26. CGF'22,24. IEEE CG&A'20.
TPAMI'19. SCF'19. UIST'24.

2018 | **Graduate Student Recruiting Committee**
Student Representative of Computer Graphics.

2016 – 2017 | **AMORE Seminar**
Imager Lab seminar organizer.

References

Alec Jacobson (Postdoc supervisor), `jacobson@cs.toronto.edu`, Associate Professor, University of Toronto.
Alla Sheffer (Ph.D. supervisor), `sheffa@cs.ubc.ca`, Professor, University of British Columbia.
Aaron Hertzmann (Internship mentor), `hertzman@dgp.toronto.edu`, Principal Scientist, Adobe Research.
Yotam Gingold (Arm's-length external evaluator), `ygingold@gmu.edu`, Associate Professor, George Mason University.